

MAT 380

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Exercise 5.2.10 (Slightly Modified)

A familiar mantra from calculus is that a differentiable function is increasing if its derivative is positive, but this statement requires some sharpening in order to be completely accurate.

(a)

Show that the function

$$g(x) = \begin{cases} \frac{x}{2} + x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is differentiable on $\mathbb{R}$ and satisfies $g'(0) > 0$.

Now prove that g is not increasing over any open interval containing 0, that is, show for any $\varepsilon > 0$, there exist $x, y \in V_\varepsilon(0)$ such that $x < y$ but $g(x) \geq g(y)$.

Proof. Let

$$g(x) = \begin{cases} \frac{x}{2} + x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Let $f_1(x) = x/2$, $f_2(x) = x^2$, $f_3(x) = \sin(x)$, and $f_4(x) = 1/x$. All of these functions are differentiable on $\mathbb{R} \setminus \{0\}$.

Then $(f_3 \circ f_4)(x) = \sin(1/x)$ is differentiable on $\mathbb{R} \setminus \{0\}$ by the Chain Rule.

The sums and products with f_1 and f_2 are differentiable on $\mathbb{R} \setminus \{0\}$ by the Algebraic Differentiability Theorem.

Thus, g is differentiable on $\mathbb{R} \setminus \{0\}$. Now at 0,

$$g'(0) = \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x} = \lim_{x \rightarrow 0} \left(\frac{1}{2} + x \sin(1/x) \right) = \frac{1}{2}.$$

Hence g is differentiable on $\mathbb{R}$ and $g'(0) = \frac{1}{2} > 0$.

Let $\varepsilon > 0$ be arbitrary. Then

$$g'(x) = \begin{cases} \frac{1}{2} - \cos(1/x) + 2x \sin(1/x) & \text{if } x \neq 0, \\ \frac{1}{2} & \text{if } x = 0. \end{cases}$$

If $|x| < 0.1$, then $|2x \sin(1/x)| < 0.2$, while $\cos(1/x)$ oscillates between -1 and 1 .

Choose sequences $a_n = \frac{1}{2n\pi}$ and $b_n = \frac{1}{(2n-1)\pi}$ with n large enough so that $a_n, b_n \in V_\varepsilon(0)$ and $a_n < b_n$. Then

$$g'(a_n) = 1.5 + 2a_n \sin(1/a_n) > 0, \quad g'(b_n) = -0.5 + 2b_n \sin(1/b_n) < 0.$$

Then $g'(a_n) > 0 > g'(b_n)$.

$\therefore g$ is not increasing over any open interval containing 0 . $\square$

(b)

Why doesn't this work for the similar function below?

$$g(x) = \begin{cases} 1.05x + x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Its derivative is

$$g'(x) = \begin{cases} 1.05 - \cos(1/x) + 2x \sin(1/x) & \text{if } x \neq 0 \\ 1.05 & \text{if } x = 0. \end{cases}$$

Note that $1.05 - \cos(1/x) \geq 0.05$. For sufficiently small $\varepsilon > 0$ (say $\varepsilon < 0.01$), we have $|2x \sin(1/x)| < 0.02$, so $g'(x) > 0$ for all $x \in V_\varepsilon(0)$. Hence this function *is* increasing near 0 .